

Manish Kumar

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Objective

Software Engineer with strong experience in backend development, distributed systems, and AI-powered applications. Skilled in building robust, scalable, and secure platforms using modern web technologies, cloud-native development, and LLM-based systems. Passionate about system architecture and high-performance infrastructure.

Education

BE (CSE) - Chitkara University | YEAR: 2026 | CGPA: 8.77 / 10

Work Experience

GoKloud Software Solutions Pvt Ltd (Deputed at **73Strings**, Bengaluru) | On-site

Software Developer Intern (Full Stack / **Java Spring Boot**) | Mar 2025 – Jun 2026

Key Contributions

- Migrated entire product's Java/SpringBoot services from Java 8 to Java 21, resolving compatibility challenges.
- Researched and implemented testing strategy with JUnit, Mockito, JaCoCo, Allure, Flyway, and Testcontainers.
- Diagnosed and resolved issues across microservices by tracing failures through APIs, business logic, and databases.
- Handled large-scale feature migrations across client environments by analyzing configurations and adapting workflows.
- Designed, developed various features and APIs, and optimized end-to-end using Redis caching and other techniques.
- Raised third-party library issues with DS Excel and SpreadJS support teams for faster resolution.
- Addressed critical security vulnerabilities identified through Qwiet AI.
- Worked on: RabbitMQ, Redis, MySQL, PostgreSQL, Elasticsearch, ApachePOI | Handled staging deployments using Devtron.

Technical Skills

- **Programming languages:** Java, Rust (Familiar), Python (Familiar), C, C++, TypeScript, JavaScript
- **Frameworks and Libraries:** Spring Boot, Node.js, Express.js, Fast-API, Angular and React.js (Working knowledge), Electron
- **AI / LLM Systems:** Advance RAG, AI Agents, LangChain, LangGraph, MCP, Hugging Face ecosystem, Recommendation Systems, Vector Search, Embeddings, Prompt Engineering, Semantic Search, Browser Agents
- **Databases & Caching:** PostgreSQL, MySQL, MongoDB, Cassandra, Neo4j (Graph DB), Elasticsearch, Redis, Neon (Serverless PostgreSQL), Qdrant (Vector DB), PG Vector | ORM: Hibernate, Drizzle
- **API & Communication:** RESTful APIs, GraphQL, gRPC (synchronous) | Apache Kafka, RabbitMQ (asynchronous messaging)
- **Cybersecurity & Identity:** OAuth 2.0, OIDC, SSO, JWT, secure session-based authentication, RBAC, ABAC, ACL, OWASP Top 10, XSS, CSRF, SQL/NoSQL Injection, IDOR, SSRF, authentication bypass, privilege escalation, LLM Security (Prompt Injection, Agent/Tool Misuse, MCP Integration Risks, Input Validation, Sandboxing)
- **Automation Testing:** Playwright, Puppeteer, Selenium (for browser automation) and Cucumber for BDD testing
- **Version Control:** Git (GitHub, Bitbucket)
- **Developer Tools:** Postman, Jira, SDM, Confluence
- **Cloud Platforms:** AWS (EC2, S3), Firebase FCM, Render, Vercel
- **Observability:** Datadog, Devtron

Distributed Systems & Scalability

- Personal project experience in designing and implementing API Gateway, distributed rate limiting, distributed locking, distributed transactions, DB migrations and sharding engine (Vitess), pgBouncer, sharding and partitioning strategies, caching write invalidation and eviction, cache fault tolerance for stampede, avalanche, and penetration strategies, and multithreading heavy applications.

Projects

NGINX-Rust API Gateway: Sub-ms API gateway with Rust FFI/Lua hot path (JWT auth, WAF, routing, rate limiting, circuit breaker); full UAM service (OAuth, token revocation, session management); zero-downtime config hot-reload via sidecar file-watch + ArcSwap – routing and upstream changes propagate to live workers in <6s, no NGINX restart required. Integrated Redis-backed distributed rate limiting and token revocation with token-version floor for instant session invalidation. [github](#) (OpenResty, NGINX, Lua, Rust, Node.js, React, Redis, MongoDB)

TestWeaver: Cloud-native AI browser agent for test automation. Converts natural-language scenarios into executable tests with AI-assisted selectors, runs them in parallel across multiple browsers, and provides built-in reporting with screenshots and step-by-step results. Designed with independently scalable API, browser workers, and AI reasoning services for enterprise-scale parallel execution across CI pipelines and release trains. (Node.js, Express, Playwright, Groq, LangGraph, React, Vite, Tailwind, MongoDB) [github](#)

JDWP Remote Debug Platform: Desktop Electron app + MCP server for live JVM debugging on K8s pods (breakpoints, threads, variable inspection, pod selection, port-forward via kubectl) with request-scoped safety. Enables debugging real production/staging failures directly on live pods, eliminating log-guessing loops; supports both MCP-driven AI-assisted debugging and manual UI-based workflows. (Java, Spring Boot, JDI, TypeScript, MCP, Node.js, Kubernetes, kubectl, Docker, React, Vite, Electron) [github](#)

Leadership & Strategy

- **Analytical Thinking:** Problem-solving, structured analysis, and data-driven decision making.
- **Business & Strategy:** Market research, business model analysis (Freemium, Subscription, Marketplace, Platform, SaaS, Transaction-based, Advertising-based models), and cross-functional collaboration.
- **Leadership & Collaboration:** Project leadership, team collaboration, work distribution, task prioritization, and strategic project planning.